

Q1 (20 July 2021 Shift 1)

AC voltage $V(t) = 20 \sin \Omega t$ of frequency 50 Hz is applied to a parallel plate capacitor. The separation between the plates is 2 mm and the area is 1 m^2 . The amplitude of the oscillating displacement current for the applied AC voltage is ____ [Take $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$]

(1) $21.14 \mu\text{A}$

(2) $83.37 \mu\text{A}$

(3) $27.79 \mu\text{A}$

(4) $55.58 \mu\text{A}$

Q2 (20 July 2021 Shift 1)

In an LCR series circuit, an inductor 30 mH and a resistor 1Ω are connected to an AC source of angular frequency 300 rad/s . The value of capacitance for which, the current leads the voltage by 45° is $\frac{1}{x} \times 10^{-3} \text{ F}$. Then the value of x is ____

Q3 (20 July 2021 Shift 2)

For a series LCR circuit with $R = 100 \Omega$,
 $L = 0.5 \text{ mH}$ and $C = 0.1 \text{ pF}$ connected across

$220 \text{ V} - 50 \text{ Hz}$ AC supply, the phase angle between current and supplied voltage and the nature of the circuit is :

(1) 0° , resistive circuit

(2) $\approx 90^\circ$, predominantly inductive circuit

(3) 0° , resonance circuit

(4) $\approx 90^\circ$, predominantly capacitive circuit

Q4 (20 July 2021 Shift 2)

Questions with Answer Keys

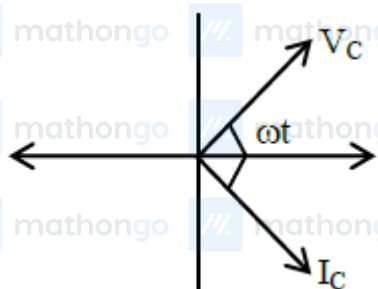
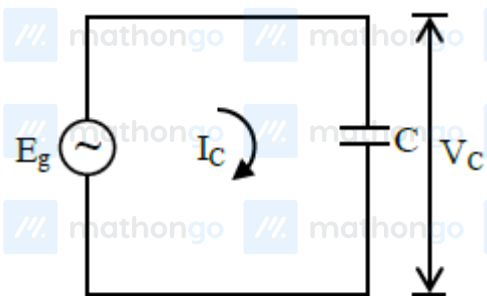
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A series LCR circuit of $R = 5\Omega$, $L = 20\text{mH}$ and

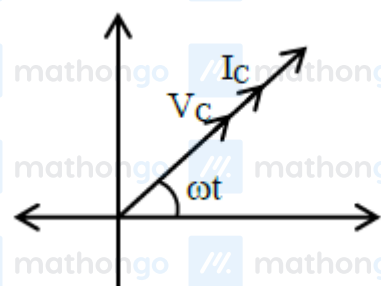
$C = 0.5\mu\text{F}$ is connected across an AC supply of 250 V, having variable frequency. The power dissipated at resonance condition is $\times 10^2$ W

Q5 (22 July 2021 Shift 1)

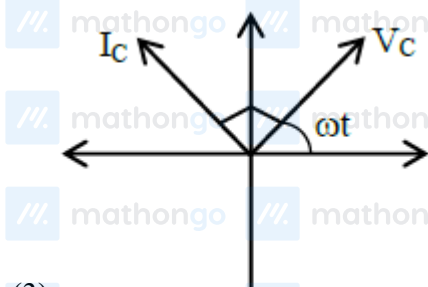
In a circuit consisting of a capacitance and a generator with alternating emf $E_g = E_{g_0} \sin \omega t$, V_C and I_C are the voltage and current. Correct phasor diagram for such circuit is:



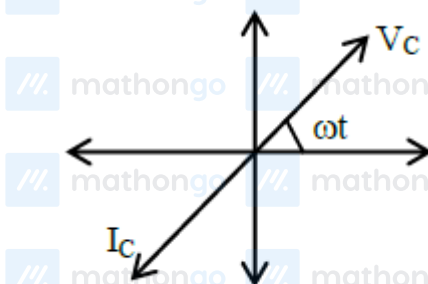
(1)



(2)



(3)



(4)

Q6 (22 July 2021 Shift 1)

Match List-I with List-II :

	List - I		List - II
(a)	$\omega L > \frac{1}{\omega C}$	(i)	Current is in phase with emf
(b)	$\omega L = \frac{1}{\omega C}$	(ii)	Current relags behind the applied emf
(c)	$\omega L < \frac{1}{\omega C}$	(iii)	Maximum current occurs
(d)	Resonant frequency	(iv)	Current leads the emf

Choose the correct answer from the options given below:

(1) (a) – (ii); (b) – (i); (c) – (iv); (d) – (iii)

(2) (a) – (ii); (b) – (i); (c) – (iii); (d) – (iv)

(3) (a) – (iii); (b) – (i); (c) – (iv); (d) – (ii)

(4) (a) – (iv); (b) – (iii); (c) – (ii); (d) – (i)

Q7 (25 July 2021 Shift 1)

Questions with Answer Keys

MathonGo

An inductor of 10mH is connected to a 20 V battery through a resistor of $10\text{k}\Omega$ and a switch. After a long time, when maximum current is set up in the circuit, the current is switched off. The current in the circuit after $1\mu\text{s}$ is $\frac{x}{100}\text{ mA}$. Then x is equal to (Take $e^{-1} = 0.37$)

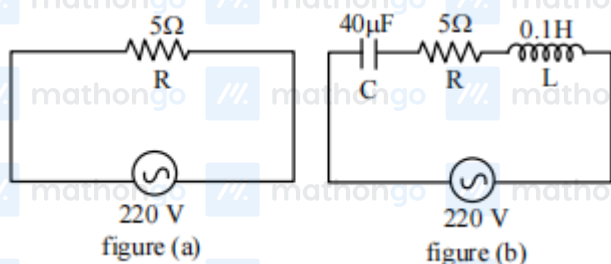
Q8 (25 July 2021 Shift 2)

A 10Ω resistance is connected across $220\text{ V} - 50\text{ Hz}$ AC supply. The time taken by the current to change from its maximum value to the rms value is:

- (1) 2.5 ms
- (2) 1.5 ms
- (3) 3.0 ms
- (4) 4.5 ms

Q9 (25 July 2021 Shift 2)

Two circuits are shown in the figure (a) & (b). At a frequency of $\omega\text{ rad/s}$ the average power dissipated in one cycle will be same in both the circuits.



Q10 (27 July 2021 Shift 1)

A 0.07H inductor and a 12Ω resistor are connected in series to a $220\text{ V}, 50\text{ Hz}$ ac source. The approximate current in the circuit and the phase angle between current and source voltage are respectively. [Take π as $\frac{22}{7}$]

- (1) 8.8 A and $\tan^{-1}\left(\frac{11}{6}\right)$

Questions with Answer Keys

MathonGo

(2) 88 A and $\tan^{-1}\left(\frac{11}{6}\right)$

(3) 0.88 A and $\tan^{-1}\left(\frac{11}{6}\right)$

(4) 8.8 A and $\tan^{-1}\left(\frac{6}{11}\right)$

Q11 (27 July 2021 Shift 2)

A 100Ω resistance, a $0.1\mu\text{F}$ capacitor and an inductor are connected in series across a 250 V supply at variable frequency. Calculate the value of inductance of inductor at which resonance will occur. Given that the resonant frequency is 60 Hz.

(1) 0.70H

(2) 70.3mH

(3) $7.03 \times 10^{-5}\text{H}$

(4) 70.3H

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Answer Key

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Q1 (3)

Q2 (3)

Q3 (4)

Q4 (125)

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Q5 (3)

Q6 (1)

Q7 (74)

Q8 (1)

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Q9 (500)

Q10 (1)

Q11 (4)

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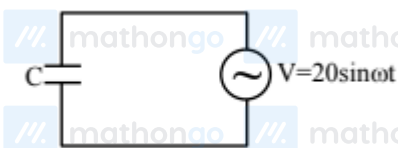
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Q1



From the given information,

$$C = \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 \times 1}{2 \times 10^{-3}} \text{ F}$$

$$\therefore X_C = \frac{1}{\omega C} = \frac{2 \times 10^{-3}}{2 \times 50\pi \times \epsilon_0} = \frac{2 \times 10^{-3}}{25 \times 4\pi \epsilon_0} \Omega$$

$$\therefore X_C = \frac{2 \times 10^{-3}}{25} \times 9 \times 10^9 = \frac{18}{25} \times 10^6 \Omega$$

$$\therefore i_0 = \frac{V_0}{X_C} = \frac{20 \times 25}{18} \times 10^{-6} \text{ A} = 27.47 \mu\text{A}$$

The value of amplitude of displacement current will be same as value of amplitude of conventional current.

Hence option 3.

Q2

$$\tan \phi = \frac{x_C - x_L}{R}$$

$$\tan 45 = \frac{x_C - x_L}{R}$$

$$x_C - x_L = R$$

$$\frac{1}{\omega C} - \omega L = R$$

$$\frac{1}{\omega C} - 300 \times 0.03 = 1$$

$$\frac{1}{\omega C} = 10$$

$$C = \frac{1}{10\omega} = \frac{1}{10 \times 300}$$

$$C = \frac{1}{3} \times 10^{-3}$$

$$X = 3$$

Q3

$$R = 100\Omega$$

$$X_L = \omega L = 50\pi \times 10^{-3}$$

$$X_C = \frac{1}{\omega C} = \frac{10^{11}}{100\pi}$$

Hints and Solutions

MathonGo

$$X_C \gg X_L$$

$$\&|X_C - X_L| \gg R$$

Q4

$$X_L = X_C \text{ (due to resonance)}$$

$$Z = R \text{ so } i_{\text{ms}} = \frac{V}{Z} = \frac{V}{R}$$

$$\frac{V^2}{R} = \frac{250 \times 250}{5} = 125 \times 10^2 \text{ W}$$

Q5

In capacitor, current lead voltage by $\frac{\pi}{2}$

Q6

(a) For $x_L > x_C$, voltage leads the current

(ii)

(b) For $x_L = x_C$, voltage & current are in same phase

(i)

(c) For $x_L < x_C$, current leads the voltage

(iv)

(d) For resonant frequency $x_L = x_C$, current is maximum

(iii)

Q7

$$I_{\text{max}} = \frac{V}{R} = \frac{20 \text{ V}}{10 \text{ K}\Omega} = 2 \text{ mA}$$

For LR - decay circuit $I = I_{\text{max}} e^{-Rt/L}$

$$\frac{-10 \times 10^3 \times x | 0^+}{4}$$

$$I = 2mAe$$

$$I = 2 \text{ mA} e^{-1}$$

$$I = 2 \times 0.37 \text{ mA}$$

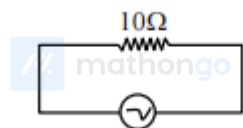
Hints and Solutions

MathonGo

$$I = \frac{74}{100} \text{ mA}$$

$$x = 74$$

Q8



$$10\Omega$$

$$V = 220 \text{ V}/50 \text{ Hz}$$

$$\Rightarrow i = i_0 \sin \omega t$$

$$\text{When } i = i_0$$

$$i_0 = i_0 \sin \omega t_1 \Rightarrow \omega t_1 = \frac{\pi}{2}$$

$$\text{When } i = \frac{i_0}{\sqrt{2}}$$

$$\frac{i_0}{\sqrt{2}} = i_0 \sin \omega t_2 \Rightarrow \omega t_2 = \frac{\pi}{4}$$

Time taken by current from maximum value to rms value

$$\Rightarrow (t_1 - t_2) = \frac{\pi}{2\omega} - \frac{\pi}{4\omega} = \frac{\pi}{4\omega} = \frac{\pi}{4 \times 2\pi f}$$

$$= \frac{1}{8 \times 50}$$

$$= \frac{1}{400} \text{ sec}$$

$$= 2.5 \text{ ms}$$

Q9

For figure (a)

$$P_{\text{mg}} = \frac{v_{\text{rms}}^2}{R}$$

$$\frac{v_{\text{rms}}^2}{Z^2} \times R = \frac{v_{\text{rms}}^2}{R} \times 1$$

$$R^2 = Z^2$$

$$25 = \left(\sqrt{(x_c - x_L)^2 + 5^2} \right)^2$$

$$25 = (x_c - x_L)^2 + 25$$

$$x_c = x_L \Rightarrow \frac{1}{\omega C} = \omega L$$

Hints and Solutions

MathonGo

$$\omega^2 = \frac{1}{LC} = \frac{10^6}{0.1 \times 40}$$

$$\omega = 500$$

Q10

$$\phi = \tan^{-1}\left(\frac{X_L}{R}\right) \quad X_L = \omega L$$

$$X_L = 2 \times \frac{22}{7} \times 50 \times 0.07 = 22\Omega$$

$$\phi = \tan^{-1}\left(\frac{22}{12}\right) \quad R = 12\Omega$$

$$\phi = \tan^{-1}\left(\frac{11}{6}\right)$$

$$Z = \sqrt{X_L^2 + R^2} = 25.059$$

$$I = \frac{V}{Z} = \frac{220}{25.059} = 8.77 \text{ A}$$

Q11

$$C = 0.1\mu\text{F} = 10^{-7}\text{F}$$

$$\text{Resonant frequency} = 60 \text{ Hz} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$2\pi f_0 = \frac{1}{\sqrt{LC}} \Rightarrow L = \frac{1}{4\pi^2 f_0^2 C}$$

$$\text{by putting values } L = 70.3 \text{ Hz.}$$